

# FairLens

## AI Bias Audit Report

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Institution: Financial Institution Audit

Audit ID: FL-2026-260856

Date: May 03, 2026

Prepared by: FairLens AI Audit System

**HIGH RISK** ■■

Composite Fairness Score: 65.0/100

Powered by IBM AI Fairness 360

EU AI Act Compliant | Confidential

EU AI Act | ECOA | GDPR | FCRA

# FairLens AI Bias Audit Report

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**Date:** May 03, 2026

**Submitted by:** CSV Upload User

## Executive Summary

This audit assessed an AI-powered loan approval system for bias, discrimination, and regulatory compliance. The system received an overall risk rating of **HIGH RISK** ■ with a composite fairness score of **65.0/100**.

The model shows moderate bias concerns that should be addressed before deployment. Remediation actions are recommended within 90 days.

## Regulatory Context

This audit evaluates compliance with:

- **Equal Credit Opportunity Act (ECOA):** Prohibits discrimination in credit decisions
- **EU AI Act Article 9:** Classifies credit scoring as high-risk AI requiring strict oversight
- **Fair Credit Reporting Act (FCRA):** Regulates use of consumer credit information
- **GDPR Article 5:** Requires data minimization and privacy protection
- **Federal Reserve SR 11-7:** Model risk management guidance

## 1. Discrimination Detection Results

### ■ *Disparate Impact Analysis*

**Disparate Impact Ratio:** 0.65

**Statistical Parity Difference:** 0

**Equal Opportunity Difference:** 0

**Interpretation:**

Analyzed 3 protected attributes. Minimum DI ratio: 0.650

### ■ *Gender DI Analysis*

**Disparate Impact Ratio:** 0.97

**Group Approval Rates:**

- Female: 52.51% (reference/highest)
- Male: 50.7% (lowest)

**DI = 0.97 ■**

### ■ *Ethnicity DI Analysis*

**Disparate Impact Ratio:** 0.93

**Group Approval Rates:**

- White: 54.03% (reference/highest)
- Hispanic: 51.37%
- Asian: 51.09%
- Black: 51.01%
- Other: 50.45% (lowest)

**DI = 0.93 ■**

### ■ *Zip\_Code DI Analysis*

**Disparate Impact Ratio:** 0.65

**Group Approval Rates:**

- 10002: 61.06% (reference/highest)
- 10000: 60.64%
- 10003: 57.69%
- 10009: 53.19%
- 10006: 52.1%
- 10005: 51.52%
- 10001: 48.45%

- 10007: 44.32%
- 10004: 44.23%
- 10008: 39.77% (lowest)

**DI = 0.65 ■**

## 2. Accountability & Governance

**Governance Maturity Level:** Defined (Level 2)

**Maturity Score:** 60/100

**Risk Level:** ■ MEDIUM

### *RACI Matrix*

| Activity                  | Responsible (R)    | Accountable (A)    | Consulted (C)   | Informed (I)       |
|---------------------------|--------------------|--------------------|-----------------|--------------------|
| Model Development         | Data Science Team  | CTO                | Compliance Dept | CEO                |
| Bias Testing & Validation | Compliance Officer | CRO                | Legal Dept      | Board of Directors |
| Regulatory Approval       | CRO                | CEO                | Compliance Dept | Regulators         |
| Production Deployment     | Engineering Team   | CTO                | Risk Management | CEO                |
| Ongoing Monitoring        | Risk Control Team  | CRO                | Compliance Dept | CEO                |
| Complaint Investigation   | Compliance Officer | CRO                | Legal Dept      | Customer Service   |
| Annual Re-audit           | External Auditors  | Board of Directors | Compliance Dept | Regulators         |

## 3. Data Privacy Compliance

### *GDPR Compliance Checklist*

- Data source documented and authorized
- Data minimization principle applied
- Retention period defined
- Regular privacy impact assessment recommended
- Data subject rights procedure should be documented

## 4. Remediation Recommendations

### ■ **HIGH PRIORITY - Action Required within 90 days**

- **Finding:** Primary Protected Attribute shows discrimination (DI=0.65)
- **Action:** Remove or replace feature in next model update
- Retrain model without this feature
- Re-run bias testing after retraining
- **Timeline:** 90 days

### ■ **COMPLIANT - Continue Monitoring**

- **Finding:** Gender within acceptable range (DI=0.97)
- **Action:** Maintain quarterly review schedule
- Document compliance status
- **Timeline:** Ongoing

### ■ **COMPLIANT - Continue Monitoring**

- **Finding:** Ethnicity within acceptable range (DI=0.93)
- **Action:** Maintain quarterly review schedule
- Document compliance status
- **Timeline:** Ongoing

### ■ **HIGH PRIORITY - Action Required within 90 days**

- **Finding:** Zip\_Code shows discrimination (DI=0.65)
- **Action:** Remove or replace feature in next model update
- Retrain model without this feature
- Re-run bias testing after retraining
- **Timeline:** 90 days

## Audit Trail

**Audit ID:** FL-2026-260856

**Audit Date:** 2026-05-03 12:15:36 UTC

**Submitted By:** CSV Upload User

**Model Version:** 1.0.0

**Next Review Date:** 2026-08-03

**Composite Risk Score:** 65.0/100

**Overall Risk Level:** HIGH RISK ■

*Generated by FairLens AI Bias Audit System*

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